TECNOLÓGICO NACIONAL DE MÉXICO

INSTITUTO TECNOLÓGICO DE PUEBLA

INGENIERÍA EN TECNOLOGIAS DE LA INFORMACION Y COMUNICACIONES

MATERIA: PROGRAMACION II

UNIDAD 2

"PORTAFOLIO DE EVIDENCIAS VISUAL BASIC "

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Introducción

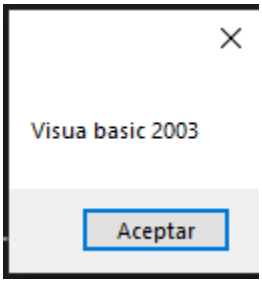
Bienvenido al portafolio de evidencias que documenta el desarrollo de proyectos en el ámbito de Visual Studio, centrándonos en la creación de programas para Windows utilizando Visual Basic en el entorno de desarrollo Visual Studio. Este portafolio es un compendio de las experiencias, habilidades y logros alcanzados a lo largo del proceso de desarrollo, destacando el dominio de herramientas tecnológicas y la capacidad para abordar desafíos de programación en un entorno Windows.

Cada proyecto presentado aquí representa un hito en el viaje de aprendizaje y desarrollo profesional, desde la concepción de la idea hasta la implementación práctica, demostrando el dominio de los principios de la programación estructurada, el diseño de interfaces de usuario intuitivas y funcionales, así como la integración de diversas funcionalidades para satisfacer las necesidades del usuario final.

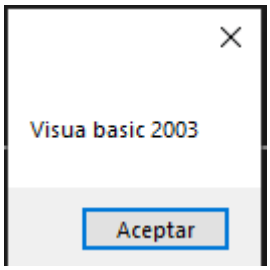
A lo largo de este portafolio, exploraremos proyectos que abarcan diferentes áreas de aplicación, desde herramientas de productividad hasta aplicaciones de entretenimiento, demostrando la versatilidad y la amplitud de conocimientos adquiridos en Visual Studio. Cada proyecto está respaldado por una descripción detallada de su funcionalidad, el proceso de desarrollo, los desafíos enfrentados y las soluciones implementadas.

Este portafolio no solo es un testimonio de habilidades técnicas, sino también un reflejo del compromiso con la excelencia, la creatividad y la innovación en el campo del desarrollo de software para Windows utilizando Visual Basic en Visual Studio. A través de la presentación de estos proyectos, se pretende no solo mostrar los resultados obtenidos, sino también inspirar a otros a explorar el emocionante mundo de la programación y el desarrollo de software.

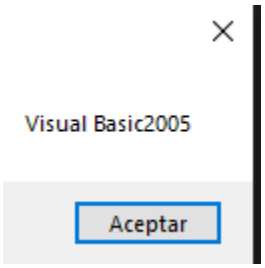
```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles
MyBase.Load
        MiMetodo()
    End Sub
    Private Sub MiMetodo()
        Dim strNombre As String
        strNombre = "Visua basic 2003"
        MessageBox.Show(strNombre)
    End Sub

End Class
```




```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles
MyBase.Load
        MiMetodo("Visua basic 2003")
    End Sub
    Private Sub MiMetodo(ByVal strTxt As String)
        MessageBox.Show(strTxt)
    End Sub

End Class
```

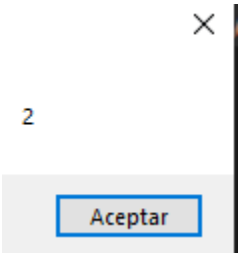


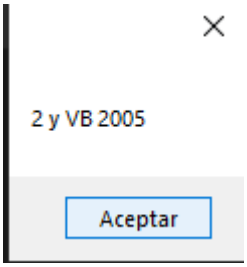

```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e As EventArgs)
Handles MyBase.Load
        Dim strValor As String = "Visual Basic"
        MiMetodo(strValor)
        MessageBox.Show(strValor)
    End Sub
    Private Sub MiMetodo(ByVal strTxt As String)
        strTxt += "2005"
        MessageBox.Show(strTxt)
    End Sub

End Class
```



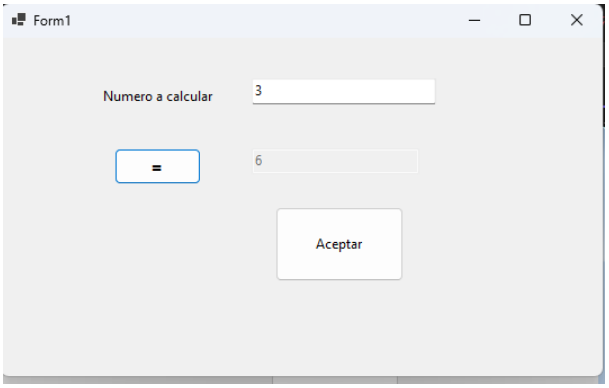

```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles
MyBase.Load
        MiMetodo(2)
```

```
        MiMetodo(2, "VB 2005")
    End Sub
    Private Sub MiMetodo(ByVal intVal As Integer, Optional ByVal strVal As
String = "")
        MessageBox.Show(IIf(strVal.Trim <> "", intVal & " y " & strVal,
intVal))

    End Sub


End Class
```

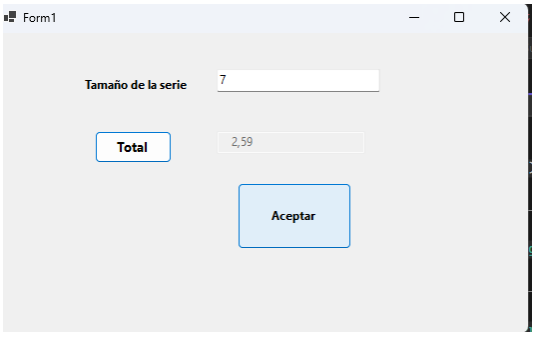




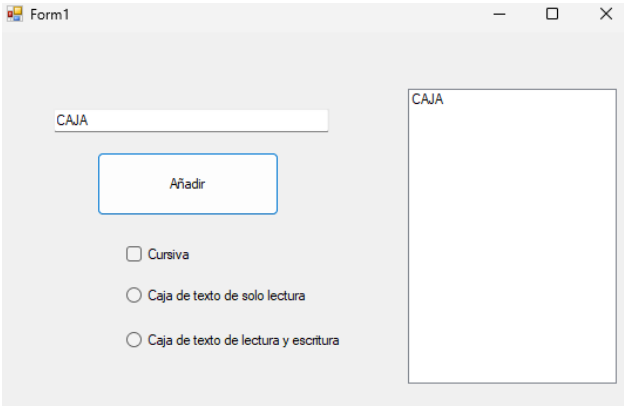

```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles
MyBase.Load

    End Sub

    Private Sub Label1_Click(sender As Object, e As EventArgs) Handles
Label1.Click

    End Sub

    Private Sub Numero_KeyPress(sender As
Object, e As KeyPressEventArgs) Handles
Numero.KeyPress
        Numero = CType(sender, TextBox)

    End Sub

    Private Sub Button2_Click(sender As
Object, e As EventArgs) Handles
Button2.Click
        Me.Close()

    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles
Button1.Click
        Dim num As Integer
        Dim i As Integer
        Dim fact As Integer = 1
        num = Convert.ToInt32(Numero.Text)
        For i = 1 To num
```



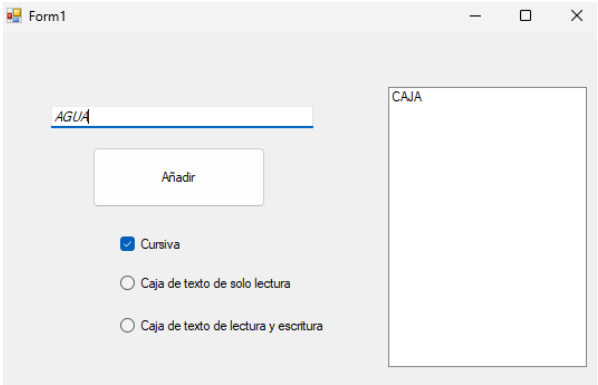
```
            fact *= i
        Next
        Resultado.Text = String.Format(fact)

    End Sub
End Class
```




```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e
As EventArgs) Handles MyBase.Load

    End Sub

    Private Sub Label1_Click(sender As Object,
e As EventArgs) Handles Label1.Click

    End Sub

    Private Sub Numero_KeyPress(sender As
Object, e As KeyPressEventArgs) Handles
Numero.KeyPress
        Numero = CType(sender, TextBox)

    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles
Button2.Click
        Me.Close()

    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles
Button1.Click
        Dim num As Integer
        Dim i As Integer
        Dim suma As Double = 0
        num = Convert.ToInt32(Numero.Text)
        For i = 1 To num
            suma += 1 / i
        Next
        Resultado.Text = String.Format("{0,8:f2}", suma)



    End Sub
End Class
```



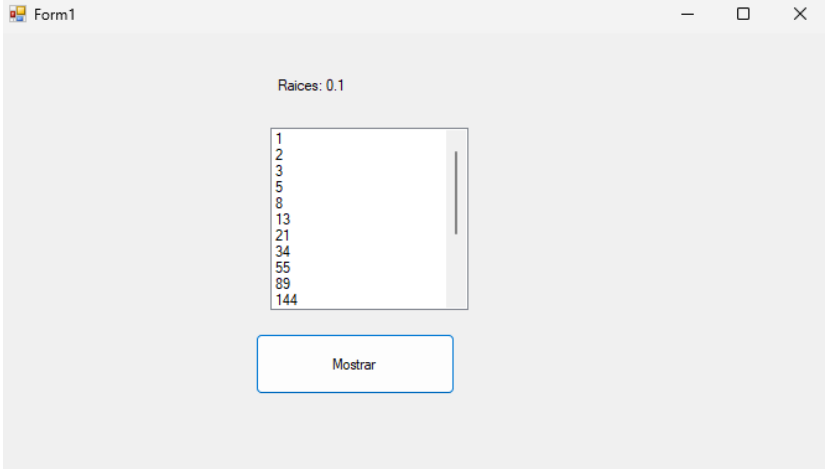

```
Public Class Form1
    Private Sub Form1_Load(sender As Object, e
As EventArgs)

    End Sub

    Private Sub botonAñadir_Click(sender As
Object, e As EventArgs) Handles
botonAñadir.Click
        If (controlDato.Text <> "") Then
Lista.Items.Add(controlDato.Text)

    End Sub

    Private Sub
casillaCursiva_CheckedChanged(sender As Object, e
As EventArgs)  Handles
casillaCursiva.CheckedChanged
        If (casillaCursiva.Checked) Then

            controlDato.Font = New Font("Macrosoft
Sans Serif ", 8.25!, FontStyle.Italic)
            Else
            controlDato.Font = New Font("Macrosoft
Sans Serif ", 8.25!, FontStyle.Regular)

        End If
    End Sub
```



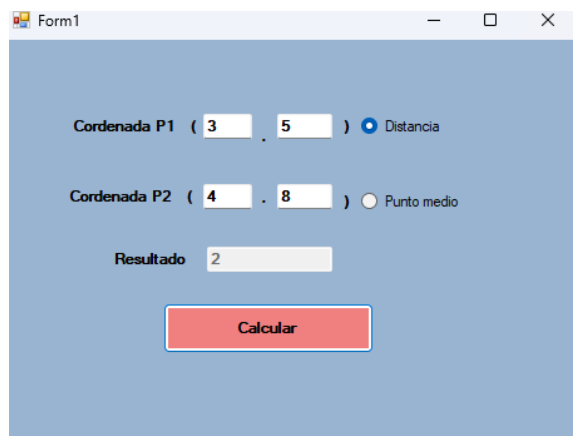

```
    Private Sub botonLectura_CheckedChanged(sender As Object, e As EventArgs)
Handles botonLectura.CheckedChanged
        If (botonLectura.Checked) Then controlDato.ReadOnly = True
    End Sub

    Private Sub botonEscritura_CheckedChanged(sender As Object, e As EventArgs)
Handles botonEscritura.CheckedChanged
        If (botonLectura.Checked) Then controlDato.ReadOnly = False
    End Sub

    Private Sub Lista_SelectedIndexChanged(sender As Object, e As EventArgs) Handles
Lista.SelectedIndexChanged
        controlDato.Text = Lista.SelectedItem
    End Sub
End Class
```

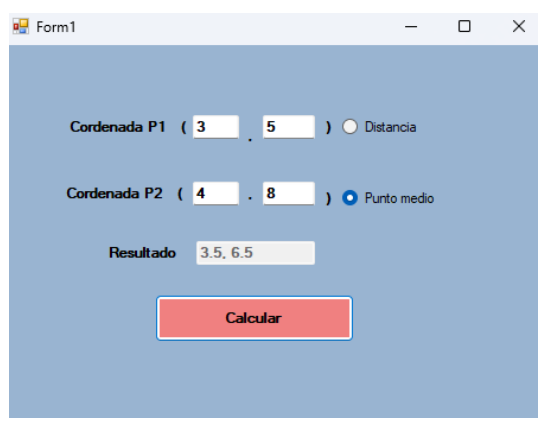
```
Public Class Form1
```

```
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles
Button1.Click
        Dim fib As Integer
        Dim anterior As Integer = 0
        Dim actual As Integer = 1
        Dim i As Integer
        Dim n As Integer = 20
        For i = 1 To n - 2
            fib = anterior +
actual
            Lista.Items.Add(fib)
            anterior = actual
            actual = fib


        Next

    End Sub
End Class
```

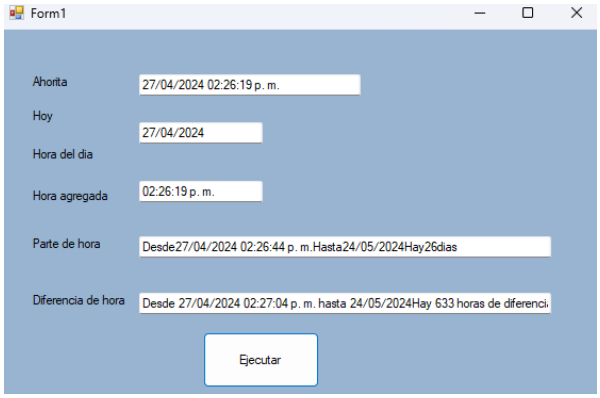



```
Public Class Form1
    Private Sub
TextBox1_TextChanged(sender As Object, e
As EventArgs) Handles x1.TextChanged

    End Sub

    Private Sub Button1_Click(sender As
Object, e As EventArgs) Handles
Button1.Click
        Dim dist As Double
        Dim punt1 As Double
        Dim punt2 As Double
        If Distancia.Checked Then
            dist =
Math.Sqrt((CInt(x2.Text) - CInt(x1.Text) ^ 2 +
(CInt(y2.Text)) - CInt(y1.Text)) ^ 2)
            Resultado.Text = dist
        ElseIf Puntomedio.Checked Then
            punt1 = (CInt(x2.Text) + CInt(x1.Text))
/ 2
            punt2 = (CInt(y2.Text) + CInt(y1.Text))
/ 2
            Resultado.Text = punt1.ToString() & ", "
& punt2.ToString()
        End If
    End Sub

End Class
```





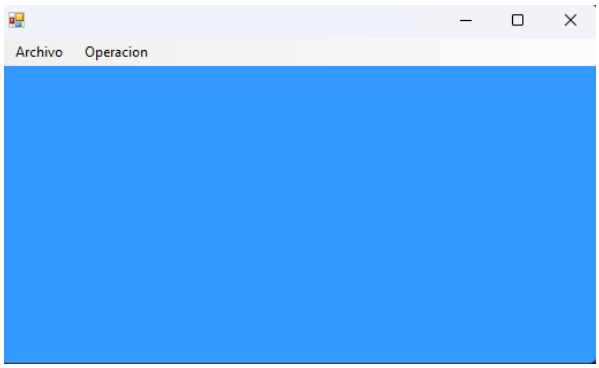

```
Public Class Form1
```

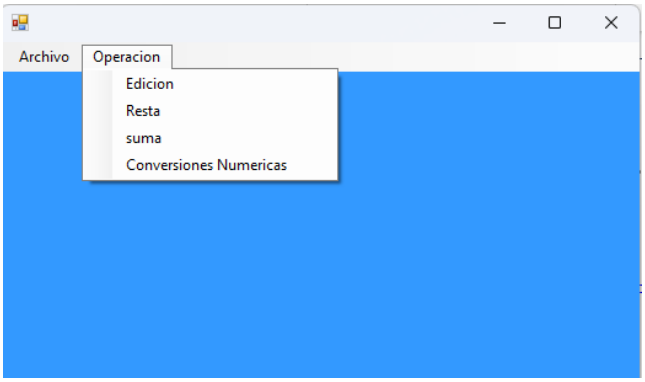



```
    Private Sub Label6_Click(sender As
Object, e As EventArgs) Handles Label6.Click

    End Sub
    Sub _Datedd()
        Dim numero, cad, FechaInicio As
String
        Dim Meses As Double
        Dim FechaFinal As Date
        Dim Intervalofecha As DateInterval
        Intervalofecha = DateInterval.Month
        FechaInicio = InputBox("Ingrese la
fecha dd/mm/aa", "Funcion DateAdd")
        FechaFinal = CDate(FechaInicio)
        numero = InputBox("Ingrese el numero de meses que se agrega a la fecha : ",
"meses")
        Meses = Val(numero)
        cad = "fecha inivio" & FechaInicio & "Meses a sumar:" & numero & " , Nueva
fecha:"
        LbLDatePart.Text = cad & DateAdd(Intervalofecha, Meses, FechaFinal)
    End Sub
    Sub _DatePart()
        Dim cad$
        Dim fechaInicio As Date = Now
        Dim fechafinal
        fechafinal = InputBox("Ingrese la fecha dd/mm/aa", "Funcion Date Part")
        LbLDatePart.text = "Desde" & fechaInicio & "Hasta" & fechafinal & "Hay" &
            DateDiff(DateInterval.Day, Now, fechafinal) & "dias"
    End Sub
    Sub _DateDiff()
        Dim fechaInicio As Date = Now
        Dim fechaFinal As Date
        fechaFinal = CDate(InputBox("Ingrese la fecha y hora en formato dd/mm/aa
hh:mm:ss", "Funcion Date Part"))
        Dim diferenciaEnHoras As Long = DateDiff(DateInterval.Hour, fechaInicio,
fechaFinal)

        LBLDateDiff.Text = "Desde " & fechaInicio & " hasta " & fechaFinal & vbCrLf
&
                           "Hay " & diferenciaEnHoras & " horas de diferencia."


    End Sub
    Sub _funcionesFecha()
        LbLNow.Text = Now
        LbLToday.Text = Today
        LbLTimeofDay.Text = TimeOfDay
        _Datedd()
        _DatePart()
        _DateDiff()
    End Sub
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles
Button1.Click
        _FuncionesFecha()
    End Sub
End Class
```

```
Public Class Form1
    Private Sub Button2_Click(sender As Object, e
As EventArgs) Handles Button2.Click
        Me.Close()
    End Sub

    Private Sub Button1_Click(sender As Object, e
As EventArgs) Handles Button1.Click
        Dim fechaNacimiento As Date
        Dim fechaActual As Date

        ' Intenta convertir los valores de los
campos de entrada en fechas
        If Date.TryParseExact(añon.Text & "/" &
mesn.Text & "/" & dian.Text, "yyyy/MM/dd", Nothing,
Globalization.DateTimeStyles.None, fechaNacimiento) AndAlso
            Date.TryParseExact(añoa.Text & "/" & mesa.Text & "/" & diaa.Text,
"yyyy/MM/dd", Nothing, Globalization.DateTimeStyles.None, fechaActual) Then

            ' Calcula la edad
            Dim edad As TimeSpan = fechaActual - fechaNacimiento

            ' Muestra la edad en años, meses y días
            MostrarE.Text = String.Format("{0} años, {1} meses y {2} días",
edad.TotalDays \ 365, (edad.TotalDays Mod 365) \ 30, (edad.TotalDays Mod 365) Mod
30)
        Else
            ' Muestra un mensaje de error si las fechas ingresadas son inválidas
            MostrarE.Text = "Fecha de nacimiento o fecha actual inválida."
        End If
    End Sub

    ' Los siguientes métodos KeyPress impiden que el usuario ingrese caracteres que
no sean números en los campos de entrada
    Private Sub dian_KeyPress(sender As Object, e As KeyPressEventArgs) Handles
dian.KeyPress, diaa.KeyPress
        If Not Char.IsControl(e.KeyChar) AndAlso Not Char.IsDigit(e.KeyChar) Then
            e.Handled = True
        End If
    End Sub

    Private Sub mesn_KeyPress(sender As Object, e As KeyPressEventArgs) Handles
mesn.KeyPress, mesa.KeyPress
        If Not Char.IsControl(e.KeyChar) AndAlso Not Char.IsDigit(e.KeyChar) Then
            e.Handled = True
        End If
    End Sub

    Private Sub añon_KeyPress(sender As Object, e As KeyPressEventArgs) Handles
añon.KeyPress, añoa.KeyPress
        If Not Char.IsControl(e.KeyChar) AndAlso Not Char.IsDigit(e.KeyChar) Then
            e.Handled = True
        End If
    End Sub
End Class
```

----------EN ESTE SE CREARON VARIOS FORMULARIOS ----------

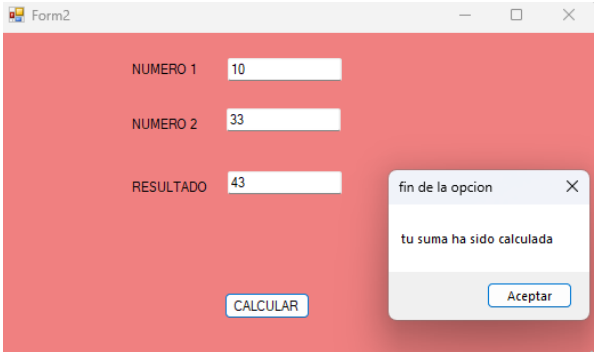





```
Public Class Form1
    Private Sub
JjdToolStripMenuItem_Click(sender As Object, e
As EventArgs)

    End Sub

    Private Sub
SumaToolStripMenuItem_Click(sender As Object,
e As EventArgs) Handles
SumaToolStripMenuItem.Click
        Dim x = New Form2
        x.ShowDialog()
    End Sub

    Private Sub
EdicionToolStripMenuItem_Click(sender As
Object, e As EventArgs) Handles
EdicionToolStripMenuItem.Click

    End Sub

    Private Sub
RestaToolStripMenuItem_Click(sender As
Object, e As EventArgs) Handles
RestaToolStripMenuItem.Click
        Dim x = New Form3
        x.ShowDialog()
    End Sub

    Private Sub ConversionesNumericasToolStripMenuItem_Click(sender As Object, e As
EventArgs) Handles ConversionesNumericasToolStripMenuItem.Click
        Dim x = New Form4
        x.ShowDialog()
    End Sub

    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load

    End Sub
End Class
```

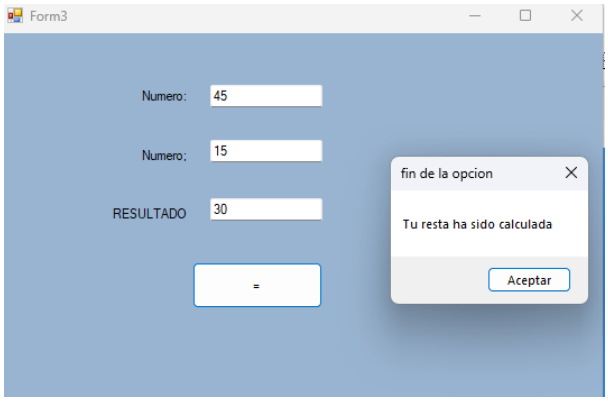
SUMA:

```
Public Class Form2
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles
Button1.Click
        Dim n1 As Integer
        Dim n2 As Integer
        Dim r As Integer
        n1 = ctNum1.Text
        n2 = ctNum2.Text
        r = n1 + n2
        ctNum3.Text = CType(r, String)
        MessageBox.Show("tu suma ha sido
calculada", "fin de la opcion ",
MessageBoxButtons.OK)


    End Sub
End Class
```



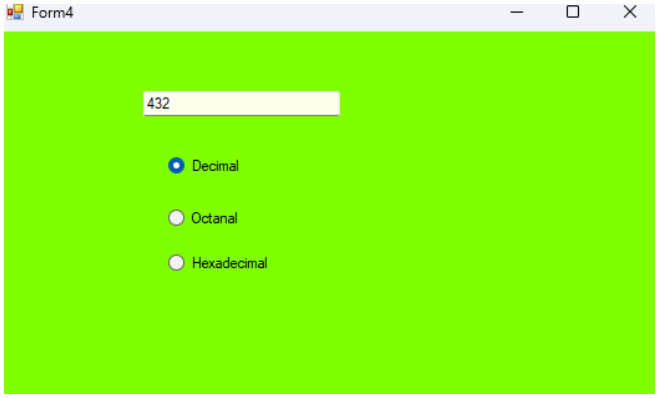
RESTA:

```
Public Class Form3
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles
Button1.Click
        Dim n1 As Integer
        Dim n2 As Integer
        Dim r As Integer
        n1 = ctNum1.Text
        n2 = ctNum2.Text
        r = n1 - n2
        ctNum3.Text = CType(r, String)
        MessageBox.Show("Tu resta ha
sido calculada", "fin de la opcion ",
MessageBoxButtons.OK)
    End Sub
End Class
```

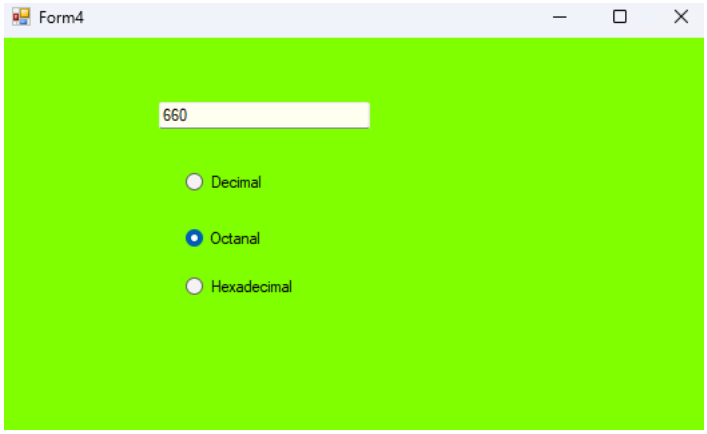


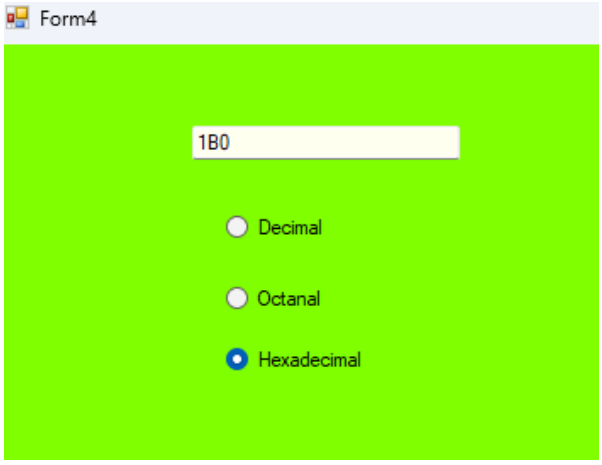
COVERCION DE NUMEROS:

```
Public Class Form4
    Public Property numeroActual As Integer

    Private Sub
btopDecimal_CheckedChanged(sender As Object,
e As EventArgs) Handles
btopDecimal.CheckedChanged
        If (btopDecimal.Checked) Then
            ctDato.Text =
Convert.ToString(numeroActual, 10)
        End If
    End Sub
```



```
    Private Sub btopOctal_CheckedChanged(sender As Object, e As EventArgs) Handles
btopOctal.CheckedChanged
        If (btopOctal.Checked) Then
            ctDato.Text = Convert.ToString(numeroActual, 8)
        End If
    End Sub

    Private Sub
btopHex_CheckedChanged(sender As
Object, e As EventArgs) Handles
btopHex.CheckedChanged
        If (btopHex.Checked) Then
            ctDato.Text =
Convert.ToString(numeroActual,
16).ToUpper
        End If
    End Sub

    Private Sub
ctDato_TextChanged(sender As Object, e
As EventArgs) Handles ctDato.TextChanged
        If (ctDato.Text.Length = 0) Then Return
        Dim texto As String = ctDato.Text
        Dim base As Integer = 0
        If (btopDecimal.Checked) Then
            base = 10
        ElseIf (btopOctal.Checked) Then
            base = 8
        ElseIf (btopHex.Checked) Then
            base = 16
        End If
        Try
            numeroActual =
Convert.ToInt32(texto, base)
        Catch ex As Exception
            If (texto = "-" Or texto = "+") Then
Return
            MsgBox(ex.Message)
        End Try
    End Sub
End Class
```





Conclusión

El uso de Visual Basic en Visual Studio ha sido una experiencia gratificante. A través de cada línea de código escrita, cada interfaz de usuario diseñada y cada función implementada, he visto cómo mis habilidades han evolucionado y mi comprensión del desarrollo de software ha crecido. Visual Studio se ha convertido en mi aliado confiable, proporcionándome las herramientas necesarias para dar vida a mis ideas y convertirlas en programas funcionales y útiles para el entorno Windows.

Cada proyecto ha sido un desafío único, pero también una oportunidad de aprendizaje invaluable. Desde la resolución de problemas hasta la optimización del rendimiento, cada obstáculo superado ha sido una lección que me ha permitido crecer como desarrollador y como persona.